



**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2022

**TECHNICAL SCIENCES P2/
TEGNIJSE WETENSKAPPE V2
MARKING GUIDELINE/NASIENRIGLYN**

MARKS/PUNTE: 75

This marking guideline consists of 7 pages. /
Hierdie nasienriglyn bestaan uit 7 bladsye.

QUESTION/VRAAG 1

- | | | |
|-----|------|-------------|
| 1.1 | A ✓✓ | (2) |
| 1.2 | D ✓✓ | (2) |
| 1.3 | A ✓✓ | (2) |
| 1.4 | A ✓✓ | (2) |
| 1.5 | C ✓✓ | (2) |
| | | [10] |

QUESTION/VRAAG 2

- | | | |
|-----|--|------------|
| 2.1 | Law of Conservation of Heat/ <i>Wet van die behoud van Hitte</i> ✓ | (1) |
| 2.2 | Specific Heat capacity/ <i>Spesifieke Warmtekapasiteit</i> ✓ | (1) |
| 2.3 | Electrolysis/ <i>Elektrolise</i> ✓ | (1) |
| 2.4 | Surrounding/ <i>Omliggende</i> ✓ | (1) |
| | | [4] |

QUESTION/VRAAG 3

3.1 If heat energy (ΔQ) is given to a system, it is used in two ways:

- (i) in increasing the internal energy of the system (ΔU) ✓
- (ii) in doing work against external pressure (ΔW) ✓

As *hitte-energie* (ΔQ) aan 'n stelsel gegee word, word dit op twee maniere gebruik:

- (i) om die interne energie van die stelsel (ΔU) te verhoog ✓
- (ii) om teen eksterne druk te werk (ΔW) ✓

(2)

3.2 An *isolated system* is a system which is not influenced by its surroundings. ✓✓

A *closed system* is a system which can **only** exchange energy but not matter with the surroundings. ✓✓

'n *Geïsoleerde sisteem* is 'n sisteem wat nie deur die omgewing beïnvloed word nie. ✓✓

'n *Geslote sisteem* is 'n sisteem wat **slegs** energie kan uitruil maar nie materie met die omgewing nie. ✓✓

(4)

3.3 Temperature/Temperatuur ✓

Pressure/Druk ✓

Volume ✓

(3)

3.4 Heat capacity of a substance is the amount of heat required to increase the temperature of the whole substance by 1 °C or 1 K. ✓✓

Warmtekapasiteit van 'n stof is die hoeveelheid hitte wat benodig word om die temperatuur van die hele stof met 1 °C of 1 K te verhoog. ✓✓

(2)

3.5 $Q_{\text{lost by 250 g water}} = Q_{\text{gained by unknown mass of water}}$

$Q_{\text{verloor deur 250 g water}} = Q_{\text{wins deur onbekende massa water}}$ ✓

$mc\Delta T_{\text{lost by 250 g water}} = mc\Delta T_{\text{gained by unknown mass of water}}$

$mc\Delta T_{\text{verloor deur 250 g water}} = mc\Delta T_{\text{wins deur onbekende massa water}}$

} Any ONE/Enige EEN

$(0,25)(4\,200)(53) \checkmark = m(4\,200)(17) \checkmark$

$\therefore m = 0,77941 \text{ kg (Accept/Aanvaar 779,41 g)} \checkmark$

(4)

3.6 Heat given out = Heat taken in

Hitte afgegee = Hitte opgeneem } Any ONE/Enige EEN ✓

$c_k m_k \Delta T = c_w m_w \Delta T$

$c_k (0,12) \checkmark (50) \checkmark = (4\,200)(0,25) \checkmark (4) \checkmark$

$\therefore c_k = 700 \text{ J.kg}^{-1}.\text{K}^{-1} \checkmark$

(6)

[21]

QUESTION/VRAAG 4

- 4.1 Internal energy is the sum of the kinetic and potential energies of all the molecules of the system. ✓✓
Interne energie is die som van die kinetiese en potensiële energies van al die molekules van die stelstel. ✓✓ (2)
- 4.2 $\Delta Q = \Delta U + \Delta W$ ✓
 $650\,000 = \Delta U + 420\,000$ ✓
 $\therefore \Delta U = 230\,000\text{ J OR/OF } 230\text{ kJ}$ ✓ (3)
- 4.3 A substance that absorbs energy (heat) from the source. ✓✓
'n Stof wat energie (hitte) vanaf die bron absorbeer. ✓✓ (2)
- 4.4 Heat engine (petrol or diesel)/Hitte-enjin (petrol of diesel)
Refrigerator (Coolant)/Verkoelingsmiddele
Hair dryer/Haardroër
Lawn mower/Grassnyer
Electrical drill/Elektriese boor
- } Any 2/Enige 2 ✓✓ (2)
- [9]**

QUESTION/VRAAG 5

5.1 **Oxidising agent** is a substance that undergoes reduction. ✓✓

Reducing agent is a substance that undergoes oxidation. ✓✓

ACCEPT

Oxidising agent is a substance that gains electrons. ✓✓

Reducing agent is a substance that donates electrons. ✓✓

Oksideermiddel is 'n stof wat reduksie ondergaan. ✓✓

Reduseermiddel is 'n stof wat oksidasie ondergaan. ✓✓

AANVAAR

Oksideermiddel is 'n stof wat elektrone bykry. ✓✓

Reduseermiddel is 'n stof wat elektrone skenk. ✓✓

(4)

5.2 5.2.1 $+2 + C + 3(-2) = 0$ ✓ $\therefore C = +4$ ✓

(2)

5.2.2 $(+1) + Mn + 4(-2) = 0$ ✓ $\therefore Mn = +7$ ✓

(2)

5.3 5.3.1 Oxygen ion/ O^{-2} is oxidised. ✓✓

Suurstof-ioon/ O^{-2} word geoksideer. ✓✓

(2)

5.3.2 Magnesium ion/ Mg^{+2} is reduced. ✓✓

Magnesium-ioon/ Mg^{+2} word gereduseer. ✓✓

(2)

[12]

QUESTION/VRAAG 6

- 6.1 Electrolytic cell/*Elektrolitiese sel* ✓ (1)
- 6.2 Electrical energy is converted to chemical energy. ✓✓
OR
 This cell needs a power source/battery. ✓✓
Elektriese energie word na chemiese energie omgesit. ✓✓
OF
Hierdie sel benodig 'n kragbron/battery. ✓✓ (2)
- 6.3 **Cathode** is the electrode where reduction takes place. ✓
Anode is the electrode where oxidation takes place. ✓
Katode is die elektrode waar reduksie plaasvind. ✓
Anode is die elektrode waar oksidasie plaasvind. ✓ (2)
- 6.4 Carbon electrodes/*Koolstof elektrodes.* ✓ (1)
- 6.5 6.5.1 Elektrode **A**
 (Metallic) brown deposit around the electrode.
 (Metaalagtige) bruin neerslag word om die elektrode gevorm. ✓✓ (2)
- 6.5.2 Elektrode **B**
Bubbles are formed around the electrode.
Borrels word rondom die elektrode gevorm. ✓✓ (2)
- 6.6 6.6.1 Electrode **B** is the anode/*Elektrode B is die anode.* ✓ (1)
- 6.6.2 Electrode **A** is the cathode/*Elektrode A is die katode.* ✓ (1)
- 6.7 6.7.1 $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ ✓✓ (2)
- 6.7.2 $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ ✓✓ (2)
- 6.8 Electroplating/*Elektroplatering*
 Galvanising/*Galvanisering*
 Purification of metals/*Suiwering van metale*
 Extraction of metals/*Ekstraksie van metale*
 Preparation of chemicals/*Voorbereiding van chemikalieë* } Any/Enige 3 ✓✓✓ (3)

[19]

TOTAL/TOTAAL: 75